

Brief Review

1. Find $\frac{dy}{dx}$ if

(a) $y = x^7 - 5x^{\frac{3}{2}} + \frac{4}{x} - 3$

$$\frac{dy}{dx} = 7x^6 - \frac{15}{2}x^{\frac{1}{2}} - 4x^{-2}$$

(b) $y = \sqrt{\cos \sqrt{x}}$

$$\text{Use the Chain Rule. } \frac{dy}{dx} = \frac{1}{2}(\cos \sqrt{x})^{-\frac{1}{2}}(-\sin \sqrt{x})\left(\frac{1}{2}x^{-\frac{1}{2}}\right)$$

(c) $y = e^{4x} \tan x^2$

$$\text{Use the Product Rule. } \frac{dy}{dx} = 4e^{4x} \tan x^2 + e^{4x} \sec^2(x^2)(2x)$$

(d) $y = \frac{3x}{\sin^{-1} x}$

$$\text{Use the Quotient Rule. } \frac{dy}{dx} = \frac{3 \sin^{-1} x - \frac{3x}{\sqrt{1-x^2}}}{(\sin^{-1} x)^2}$$

(e) $x^{\frac{1}{2}} - y^{\frac{1}{3}} = x$

$$\text{Use Implicit Differentiation. } \frac{1}{2}x^{-\frac{1}{2}} - \frac{1}{3}y^{-\frac{2}{3}}\frac{dy}{dx} = 1. \text{ Solving for } \frac{dy}{dx} \text{ yields } \frac{dy}{dx} = \frac{1 - \frac{1}{2}x^{-\frac{1}{2}}}{-\frac{1}{3}y^{-\frac{2}{3}}}.$$

2. Find the maxima, minima, and points of inflection of $f(x) = x^3 + 3x^2 - 9x + 2$. Sketch the graph.

First note $f'(x) = 3x^2 + 6x - 9 = 3(x+3)(x-1)$. Thus the critical points are $x = -3, 1$. Next $f''(x) = 6x + 6 = 6(x+1)$, which is positive for $x > -1$ and negative for $x < -1$. Thus $x = -1$ is an inflection point. Moreover, the second derivative test implies that f has a max at $x = -3$ and a min at $x = 1$.

3. Evaluate the following integrals.

(a) $\int \sqrt{x} + x^{-\frac{3}{2}} dx$

$$\frac{x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}} + C$$

(b) $\int \frac{\sin x \, dx}{(1+2 \cos x)^2}$

Let $u = 1 + 2 \cos x$. Then $du = -2 \sin x$ and the integral becomes $-\frac{1}{2} \int \frac{du}{u^2} = \frac{1}{2} u^{-1} + C = \frac{1}{2} \frac{1}{1+2 \cos x} + C$.

(c) $\int \frac{e^{2x} \, dx}{3 - e^{2x}}$

Let $u = 3 - e^{2x}$. Then $du = -2e^{2x}$ and the integral becomes $-\frac{1}{2} \int \frac{du}{u} = -\frac{1}{2} \ln |u| + C = -\frac{1}{2} \ln |3 - e^{2x}| + C$.

4. Find the area under the graph of $y = 3x^2 - 1$ over $[0, 2]$.

The area is given by $\int_0^2 3x^2 - 1 \, dx = x^3 - x \Big|_0^2 = (8 - 2) - (0 - 0) = 6$.